Electrically Conductive Adhesive Interconnections on Additively Printed Substrates

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***Abstract*— In this paper, the assembly of passive components on additively printed metallization using additively dispensed conductive adhesive is studied. Additively printing technologies are being increasingly explored in the realm of electronics manufacturing. These have found their way in various aspects of manufacturing starting from achieving reliable traces to multi-layer circuits with micro-via, and now surface mount devices. Different technologies have their merits and demerits based on the conductive material properties, substrate, and the printer's capability to incorporate multiple materials simultaneously. Previously, authors have shown how Aerosol Jet Printing (AJP) technology can be utilized to deposit very fine traces and realize multi-layer metallization with additively printed donut-vias. To move on to the other aspect of electronics manufacturing, assembly of Surface Mount Devices (SMD) on a flexible substrate is being looked into using additive methods. For the assembly, having an additively printed pad along with a conductive adhesive appears to be a suitable path and an alternative to solder reflow process where components suffer high temperatures. However, certain process limitations come to light in this aspect. For example, instead of copper being used as a conducting material, silver is used due to its availability and ease in thermal curing as opposed to copper because of oxide formation. Another example could be the depositing of conductive adhesive. Since the adhesive is extremely viscous, printer should be able to incorporate high viscosity materials without any blockage issues every minute. In this work, authors have utilized two different printing technologies: InkJet and micro-dispensing, to characterize the assembly process of SMD components. InkJet technology is used to deposit the metal conductive pad, on top of which conductive adhesive is pneumatically dispensed. SMD components are placed using a semi-automatic pick-and-place machine. Different SMD discreet components are used such as resistors, capacitors, and inductors. Process studies on the two state-of-art printing technologies in achieving a fine quality print are developed. With this work, authors aim to characterize the printing processes of two materials, and also providing a viable method for the assembly of SMDs on a flexible substrate.**

Keywords*— *inkjet, direct write, printed electronics, flexible substrate, conductive adhesive, surface mount devices, satellite drops, droplet behavior

Nomenclature:

AJP	Aerosol Jet Printing
SMD	Surface Mount Detectors
CAD	Computer Aided Design
cP	Centipoise, Unit of Viscosity
IJ	Inkjet Printing
LED	Light Emitting Diode
ECA	Electrically Conductive Adhesive
CIJ	Continuous Inkjet printing
DOD	Drop-on-Demand Inkjet Printing
TIJ	Thermal Inkjet Printing
PZT	Lead Zirconate Titanate, piezo element
DXF	Drawing Exchange Format
DW	Direct Write

I. Introduction

Additive printed electronics is a rapidly expanding research area that continues to show promising results for the manufacturing of low-cost flexible electronics. The additive processes have been in development for various aspects of electronics, initially starting out from reliable traces to multi-layer substrates to surface mounted devices. The benefits of these processes far outweigh the conventional technologies that involve multi-step lithography etching as opposed to simple CAD file. Although the conventional processes are well developed for copper patterning, the plastic substrate used in flexible electronics is chemically incompatible to resists and etchants. Recently, different additive technologies are being explored to achieve the desired fine resolution with the compatibility of wide ranges of materials. The most common one being the screen-printing technology, that has been in the industry for many years, commonly used for mass manufacturing, however, cannot achieve the fine resolution due to limits on the screen mesh size. Having a multi-material print comes to be quite difficult as well. Other technologies such as

Aerosol Jet, Direct Write, InkJet, are more preferable because of their capability to incorporate multiple materials together and getting down to <10 μm resolution. Unfortunately, they have their own advantages and disadvantages, mainly in the materials department. One technology, although can have wide range of materials, but cannot do all type of materials such as pastes, inks. For example, Aerosol Jet is limited to material viscosity ranging from 1-1000 cP, Direct Write is limited to dispensing thick viscous pastes up to 100,000 cP, InkJet limited to narrow 5-30 cP materials. Whatever the limitation, each of these can offer something better than the other.

Printing techniques such as IJ, AJP, micro-dispensing have all been exploited in various industries, such as graphics, medical, textile, but their major use can be found in printed electronics. Different direct-write techniques, along with detailed reviews of IJ and AJP have been pursued by researchers [1]-[4]. The reason for the ongoing development in the field of printed electronics is the fact that the amount of waste produced is very little, in addition to allowing researchers and scientists to overcome the expensive conventional manufacturing methods. Because of this, various articles are published on printed electronics such as [5] discussing the use of IJ technology for conductive tracks, [6] quantifying the use of metal-based IJ inks, [7] reviewing the materials, processes for flexible and stretchable electronics, [8] focusing on wearable electronics. Authors of this article have also prior published extensive research on Aerosol Jet technology in which process consistency study was done to realize multi-layer substrates to additively print micro-vias [9]-[12]. Other applications where IJ has been used are organic LED displays [13], thin-film transistors [14], and thin-film batteries [15]. IJ being relatively fast technique can also be scaled up for roll-to-roll process as well making it a remarkable technology for integration into a manufacturing process. Another direct write technique, quite different from IJ, in which lot of research has been done is micro-dispensing. Instead of droplets, pastes are dispensed using a pressure via a nozzle tip. The technology has been employed in antennas [16]-[18], RFID tags [19], 3D passive harmonic transceiver [20]. Another interesting study evaluates the interconnection between micro dispensed conductive ink and traditional hardware, in which the performance of conductive epoxy and low temperature solders are quantified [21]. A similar interconnection study is also published that evaluates the joint stability from direct soldering with lead solder on inkjet and screen printed structures [21].

In terms of throughput for both of these techniques, IJ provides a dominant hand on micro-dispensing printing, owing to the fact that there are hundreds, and even thousands of nozzles used during the printing, as compared to the other. Due to multiple number of nozzles and with printing speeds in hundreds of mm/s, a complex print can be done in few minutes, which makes it a perfect technique for both, small and high volume production. However, every technique comes with its drawbacks as well. With inkjet printing, materials used can only be in liquid form within a certain range of viscosity, while micro-dispensing allows printing of pastes as well and can accommodate wider range of viscosities. Number of nozzles used for printing and speeds are much higher in IJ as compared to micro-dispensing. Clogging of the nozzles in IJ are much more frequent with little to no way of recovering them, however for micro-dispensing, the nozzle can be easily removed for cleaning. Although both the printing platforms comes with its own limitations, the benefits far outweighs their flaws. Being able to print on any type of substrate, 2D or 3D, flexible, stretchable or rigid, with just a click of a button, brings a lot to the table of flexible hybrid electronics systems.

From a quick literature review, it can be seen that not much work is done in the area of process development where one can look at the effect of print parameters on some of the physical, electrical, and mechanical properties. Print parameters often can be process specific, but include the parameters that can greatly impact the quality of the print. They also include post-print parameters such as sintering temperature and time that vastly effect the electrical and mechanical properties. In this paper, the authors use Inkjet technology in conjunction with DirectWrite to achieve additive interconnections. Inkjet is utilized to print metal conductive pads with silver ink, while DirectWrite is utilized to dispense a viscous conductive adhesive to create the interconnection for mounting the SMDs. These interconnections will be realized on a flexible substrate, polyimide. The paper is divided in two sections: First, study with Inkjet will be done to quantify the waveform, droplet characterization, effect of post-print parameters. Electrical property such as resistance, and mechanical property such as shear load to failure will be quantified with the optimized waveform parameters against various sintering temperature and time. A statistical study with control charting will also be done to quantify the process capability; Second, study will be on DirectWrite where ECA will be dispensed on optimized print parameters and SMDs will be mounted.

A. Inkjet Printing Technology

Inkjet printing is a deposition technique that additively builds up structures by generating droplets placed at a point with extreme precision and volume. The process can achieve high quality and reproducible structures, due to which it is recognized as an advanced and economical method for electronics manufacturing. The printing is further classified into two categories. Continuous inkjet (CIJ) and Drop-on-Demand (DOD). As the name suggests, in CIJ, a continuous stream of material is running through the print head that leads to a continuous flow of droplet production. As a consequence, some of the droplets gets unused and are re-circulated back into the reservoir resulting in potential contamination. To avoid this, DOD technique is introduced in which droplets are only generated when required i.e. more efficient. Although there is a drawback of jets clogging when idle, this technique is generally preferred over CIJ. Droplets generated through DOD are further classified into two types: Thermal inkjet (TIJ) and piezoelectric DOD. TIJ are low-cost printers in which a resistor inside the ink chamber is heated up and the bubble of ink vapor pushes the ink out the nozzle. This technique requires the ink to have a volatile component to form the vapor bubble. In piezoelectric DOD, a voltage pulse is supplied that actuates that piezoelectric transducer (PZT), creating pressure waves within the ink reservoir, thus generating the droplets. Among these two, the

preference is given to piezoelectric inkjet printing due to the larger extent of control over the velocity and volume of the droplet. A flowchart explaining the different categories is shown in Fig. 1.

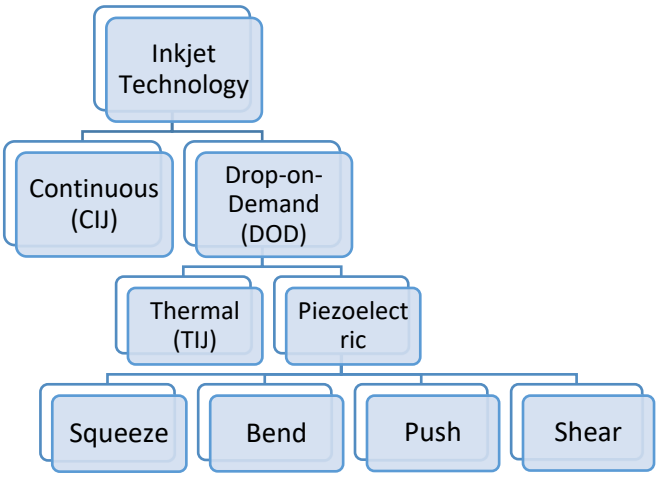


Fig. 1. Flowchart of Inkjet Technology

The piezoelectric DOD technology can also be classified in four mode types: squeeze; bend; push; and shear. For the extent of this paper, the discussion will only focus on shear mode. Fig. 2 shows the structure of shear-mode piezo actuator. The piezoelectric walls formed by piezoelectric ceramics covers the ink pressure chamber. A voltage pulse is applied to the walls that results in shear-mode deformation, due to which the ink droplets are ejected. The pulse consists of forward and reverse voltage which creates the corresponding electric field that increases and decreases the volume of the ink pressure chamber respectively. This process happens in a sequence resulting in a pressure wave inside the ink chamber that ejects the droplet out of the nozzle. The cross-section of the ink chamber is shown inset of Fig. 2. Detailed process can found in literature [23].

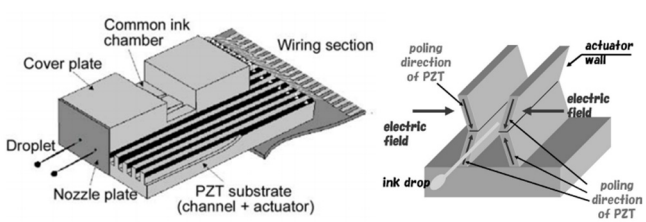


Fig. 2. Shear Mode Piezo Actuator with cross section view (right)[23]

Basics of Waveform

A waveform, in a piezo-based inkjet printer, defines how much and for how long the PZT element should deform to create a nice, clean, and round droplet. It consists of voltage that determines how much and pulse timings that determines how long. Both of these can have a great impact on the droplet shape, speed, stability, and the final print. The waveform starts with a ON voltage parameter that deforms the PZT and expands the ink channel to draw the ink followed by ON pulse width which specifies the time the ON voltage should be supplied. The waveform then moves to OFF voltage to contract the ink channel back to push the liquid and eject the droplet and OFF pulse width repeatedly for thousands of times in a second. Voltage parameter is responsible for the droplet volume and its speed, while pulse width is responsible for determining the maximum speed of the droplet. In general, ON voltage is set to be twice of OFF voltage, and OFF pulse width is set to be twice of ON pulse width. Apart from the droplet characteristics, these parameters also decide the jetting frequency of the print head that is different for every print head.

Equipment in-house

The inkjet printer used in the study is from SUSS MicroTec, PiXDRO LP50 desktop R&D inkjet printer, shown in Fig. 3. The printer is equipped with piezo-electric DOD Konica Minolta (KM) print head assembly. The print head, shown in Fig. 4, consists of 512 jetting nozzles, 256 nozzles arranged in one line, with 42pL as nominal droplet volume. Other specifications for the head are listed in TABLE I.

Fig. 3. Inkjet Printer in-house

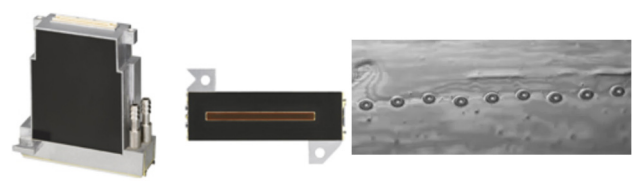


Fig. 4. Konica Minolta 512 Print Head and Nozzle Placement

TABLE I. KM512 Print Head Specifications

KM512	
Technology	Piezo Drop-on-Demand (Shared wall 3 cycle)
Resolution	180dpi × 2lines = 360dpi
Number of Nozzles	256 nozzles × 2lines = 512 nozzles
Nozzle Pitch	70.5µm
Drop Size	42 pL
Max Frequency	7.6kHz
Printing Width	36.1mm

B. Direct Write Micro-Dispensing Technology

Micro-dispensing or direct-write is another digital printing technique that does a remarkable job in additively printing wide range of materials. It does so in exceptionally small volumes with very high precision, that is achieved because of its ability

to dispense much closer to the substrate surface. Micro-dispensing is loosely translated as dispensing of material using a certain amount of pressure and pressing it out through a small orifice onto a substrate. To get into more details about the process and the technology, the discussion will be focused on one of the products from nScrypt, Inc, who have become the pioneers in this technology. The printer consists of a quad head assembly with SmartPump™ and nFD™ gizmos. The system also comes with a process view camera and laser for alignment purposes, if necessary. Since the materials used are conductible metal pastes, and considering that the nFD™ is filament fed material extrusion tool, more details on the SmartPump™ gizmo and its working are discussed next.

The SmartPump™ gizmo mainly consists of a valve rod, the most important component in the gizmo housing that controls the flow of material extrusion. It requires the material to be loaded into a syringe, with one end connected to the pressure tube and the other to the valve body. The "open" and "close" position of the valve rod defines when to let the material flow and when to stop it without any kind of leaking or drooling. This movement of valve rod happens automatically, once the positions are defined, in a coordinated manner as per the requirement of design file. These positions along with few other process parameters can have a drastic impact on the quality of the print. Some of them include print speed, material pressure, nozzle height, that need to be set as well. The benefit of not having a complicated setup with the material being well-contained in a syringe makes it a perfect additive tool for advancement of the manufacturing of 3D printed electronics. More details on this and other gizmos can be found on the company website [24].

Equipment in-house

The DW printer used in this study is from nScrypt, 3Dn-TableTop with quad-head assembly consisting of two SmartPumps™ for material dispensing and two nFD™ gizmos for filament-based material extruder. The printer with close-up view of the gizmos is shown in Fig. 5.

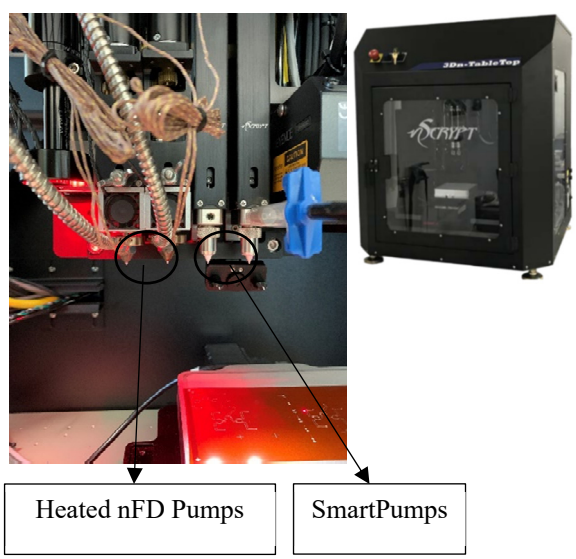


Fig. 5. Direct Write Micro-dispensing Printer in-house

II. Experimental

One of the aims of this paper is to look at the additive interconnection of SMD devices on flexible substrates. The test design was modeled keeping this is mind that involves printing of multiple SMD devices footprints such as resistors, inductors, LEDs. The modeling was done in AutoCAD software to generate appropriate files needed for both the printing systems, such as image files for IJ and DXF files for DW. Apart from resistors and inductors, a LED is also mounted for display purposes. The design is shown in Fig. 6.

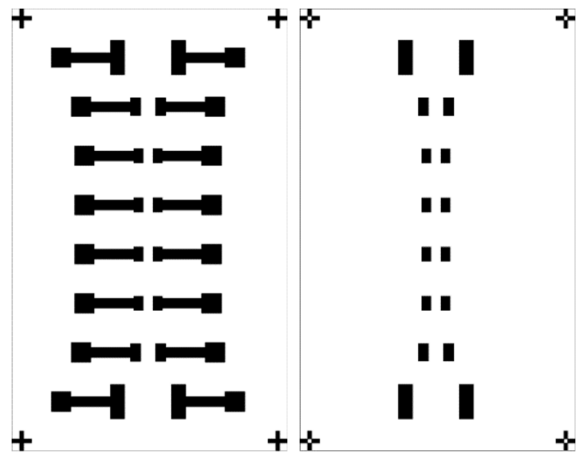


Fig. 6. Test Design for SMD Interconnections

A. Material Properties

The materials used in the paper consists of inks and pastes suitable for both, IJ and DW printing. For IJ, a Nanoparticle based silver ink is used with low volume resistivity and viscosity compatible with IJ and KM512 assembly. For DW, a silver metal conductive adhesive is used to attach the SMDs. Since DW printer is able to dispense viscous materials, the viscosity of the adhesive reflects this capability. Typical properties of both the materials are listed in TABLE II and TABLE III.

TABLE II. Typical Properties of Silver Nanoparticle Ink

Typical properties of Silver IJ Nanoparticle ink	
Physical State	Ink
Appearance	Dark gray
Mean particle size (nm)	100
Silver content (wt. %)	19
Density (g/mL)	1.3
Viscosity (cP)	12-18
Dry Film Thickness (µm)	1-2
Sintering	150°C for 30mins

TABLE III. Typical Properties of Silver DW Micro-dispensing Paste

Typical properties of Silver DW adhesive paste	
Physical State	Silver Paste
Viscosity (cP)	1 s^{-1}: 110,000; 100 s^{-1}: 15,000
Specific Gravity	3.2
Curing	140°C for 60mins Or 150°C for 30mins

B. *Procedure*

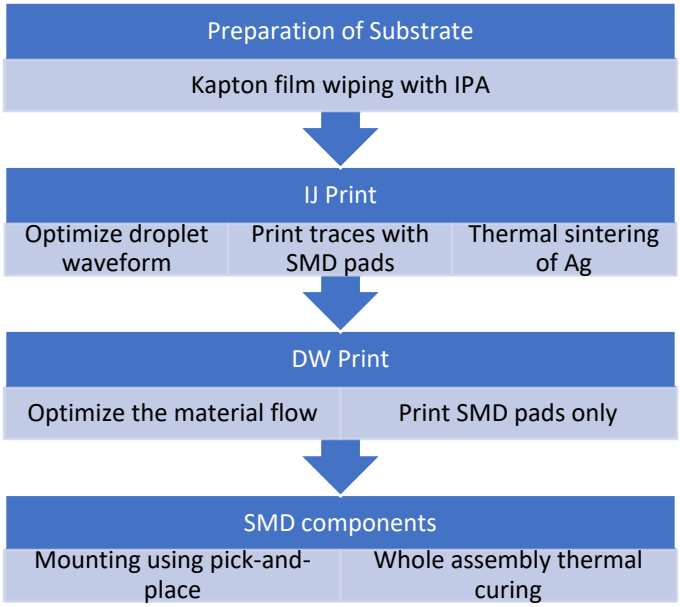


Fig. 7. Flowchart of Printing Process

Initially, a pad with silver Nano-particle ink is inkjet printed on the polyimide film, following the design shown in Fig. 6 (left). The ink is thermally sintered to evaporate the surfactant. The print is brought to DW. After alignment, a silver conductive adhesive is printed with the design shown in Fig. 6 (right). The components are mounted, and the whole assembly is cured thermally at the recommended conditions of silver ECA. All of the SMDs are mounted using a semi-automatic pick-and-place machine for proper alignment. A flowchart detailing the procedure is shown in Fig. 7.

III. Results and Discussion

Results of the process study done on both the printing technologies are presented. This section is divided into three sub-sections: IJ printing; DW printing; and mounting of SMD components. More details on these are provided next.

A. *IJ Printing*

For additive interconnection between a SMD and flexible substrate, IJ printing is done first in which trace with SMD footprint is printed on the polyimide film. Now, with any printing technique, to realize a print, one must optimize the print process parameters to obtain a fine quality print. In IJ printing, most of the work comes in determining the correct waveform parameters for stable jetting of the droplets. Basics of a waveform and some of the parameters are discussed in Introduction section. In this section, a closer look on those parameters and their effect on the jetted droplet is shown. Final selected waveform is presented along with the droplet formation. Aside from the print parameters, effect of post-print parameters such as sintering temperature and time is also studied. Electrical and mechanical properties are characterized for various temperatures and times.

Droplet Formation for Optimized Waveform

Waveform parameters were optimized for the ink used, and the droplet formation was recorded, shown in Fig. 8. This figure is recorded from the time the droplet exits the nozzle with a ligament that is shown at 50 µs up to 200 µs where a clean round droplet is formed.

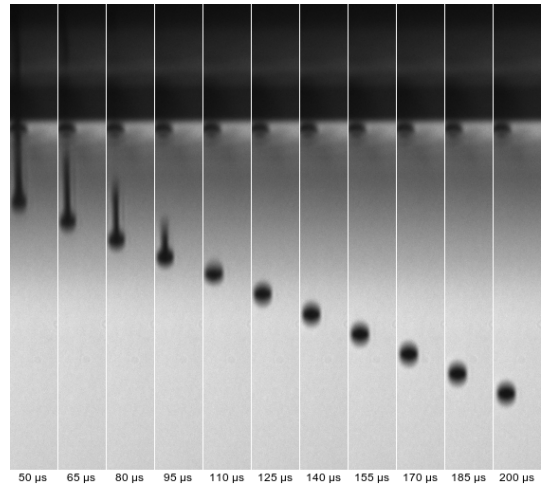


Fig. 8. Droplet Formation for Finalized Waveform

Effect of Voltage on the Droplet

Voltage parameter in an inkjet waveform can have a great impact on the droplet speed and its volume. It determines how much to deform the PZT element forcing the ink to move within and eject a droplet. It consists of two voltages: ON voltage and OFF voltage. When a ON voltage is applied, it deforms the PZT to expand the ink channel and draw the ink. The ink channel is then contracted by OFF voltage that pushes the ink out the nozzle in a form of droplet. In general, for the print head used, ON voltage is set to be twice of OFF voltage. Fig. 9 shows the effect of increase in ON voltage on the droplet formation. This is recorded at the same moment of time, that shows the droplet speed increases with the increase in voltage, up-to a point. It shows an optimal voltage value after which satellites or ligaments start to form and increase. This satellite formation, apart from bad print, can also result in circuit shortage.

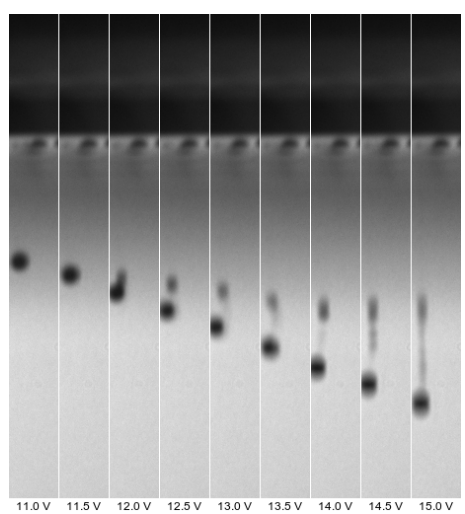


Fig. 9. Effect of Voltage on Droplet Formation

Effect of Pulse Width on the Droplet

Pulse width determines how long to deform the PZT element for jetting the droplets. When the ON voltage is supplied and the ink channel is expanding to draw the ink, the ON pulse width comes into play that specifies the time to keep supplying the ON voltage. Considering the length of ink channel and the time pressure waves take to travel back and forth, it has a great impact on the speed of droplet.

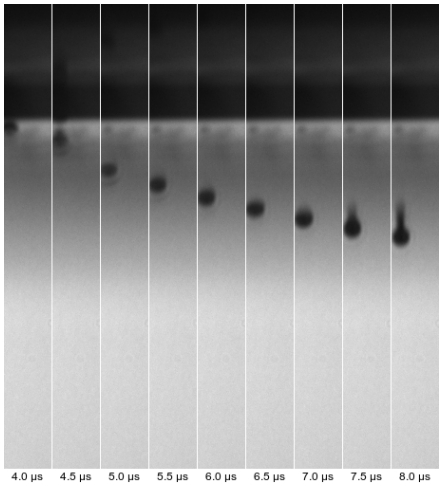


Fig. 10. Effect of Pulse Width on Droplet Formation

The pulse width consists of ON pulse width and OFF pulse width. In general, OFF pulse width is set to be twice of ON pulse width. Fig. 10 shows the effect of varying ON pulse widths on droplet formation at the same moment of time. Clearly, for a set ON voltage, higher droplet speeds are seen at higher ON pulse width. But after a certain point, here 8.0µs, the movement inside the ink channel becomes out of sync, and speed starts to decrease.

Effect of Phase Width on the Droplet

Phase width is another waveform parameter that determines the speed consistency of all the droplets. Since the print head used is a shear mode type head, only one nozzle jets at a single time. The KM512 head uses a 3-cycle drive technology that divides the nozzles into groups of 3 for timed ejection control. Between each jetting nozzle, there is a 70µs delay that results in droplet formation shown in Fig. 11. For a phase width chosen to be 43µs, the speed of the droplets appears to be consistent.

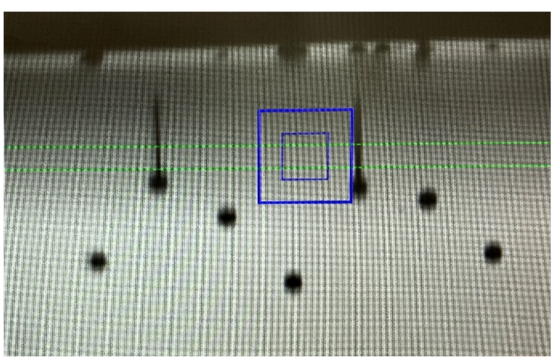


Fig. 11. Droplet Formation for Finalized Phase Width

Long Term Stability of the Droplet

Long term stability is more of a reliability parameter than waveform, that shows the stability of the droplet over certain number of runs. In Fig. 12, it is clear that the droplet formation over 40 number of runs is quite stable for the final waveform selected. In this context, the number of runs implies the number of times jetting has stopped and started again.

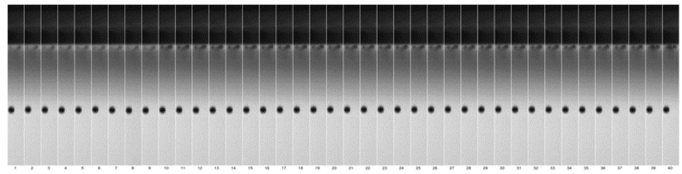


Fig. 12. Long Run Stability of Droplet

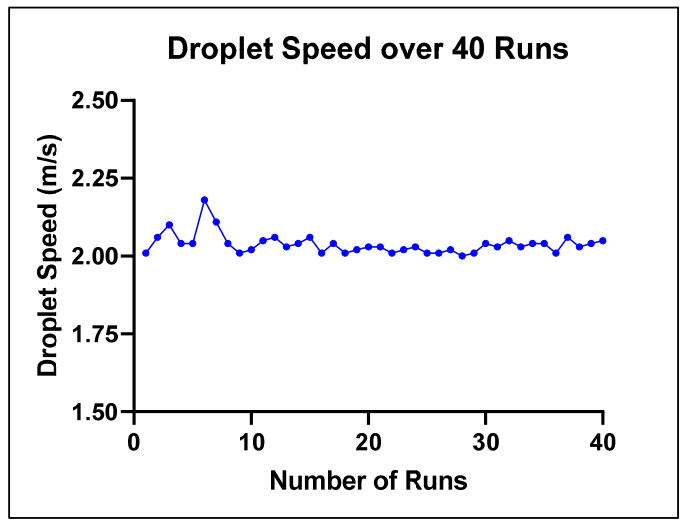


Fig. 13. Droplet Speed over 40 runs

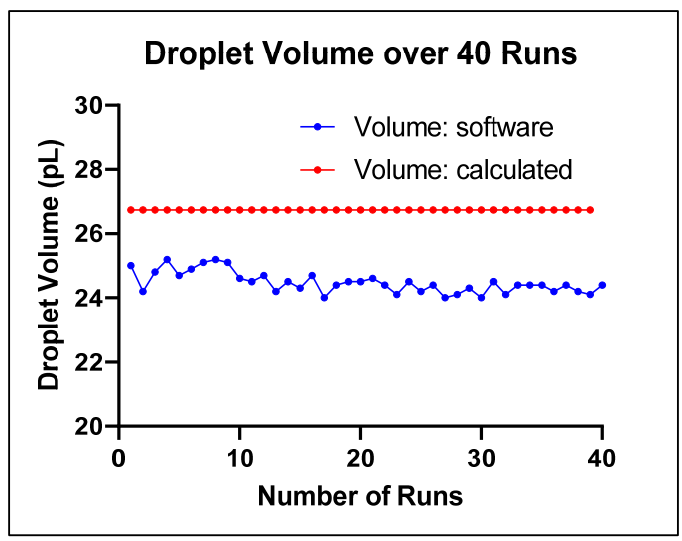


Fig. 14. Droplet Volume over 40 runs

To quantify the droplet further, speed and volume of ejected droplet was recorded during the 40 runs, and plotted to see the variability. Fig. 13 shows the droplet speed and Fig. 14 shows the droplet volume when ejected from the nozzle. Both of these parameters are recorded from the printer software,

however for droplet volume, it was manually calculated as well, knowing the pitch of the nozzle. Calculated droplet and volume recorded from the software provide a good agreement between them.

Effect of Post-Print Parameters

Post print parameters in additive printed electronics include mostly sintering conditions. Sintering temperature and time can have detrimental effects on the electrical and mechanical properties of the printed material. Different samples were printed and subjected to various temperature and times, which were selected based on the recommended sintering conditions for the ink listed in TABLE II. Fig. 15 shows the graph in which sintering time was kept constant for 30 mins, but temperature was varied. Very high resistivity was seen initially since the ink was not sintered completely, and with increase in temperature, it dropped down to a reasonable value. Multiple samples shows the repeatability of the test. Fig. 16 shows the graph in which sintering time was varied but temperature was kept constant at 150°C. Lot of variation was seen between samples when sintered for 15 mins, but it reduced when the time was increased to 30 mins, and then negligible difference was noticed.

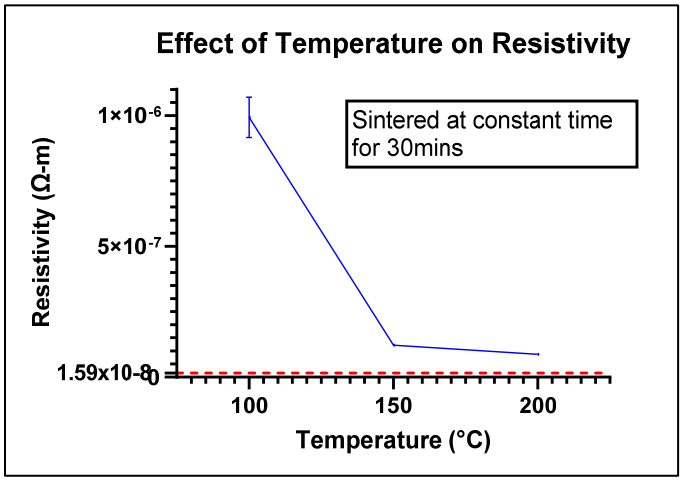


Fig. 15. Effect of Sintering Temperature on Electrical Resistance

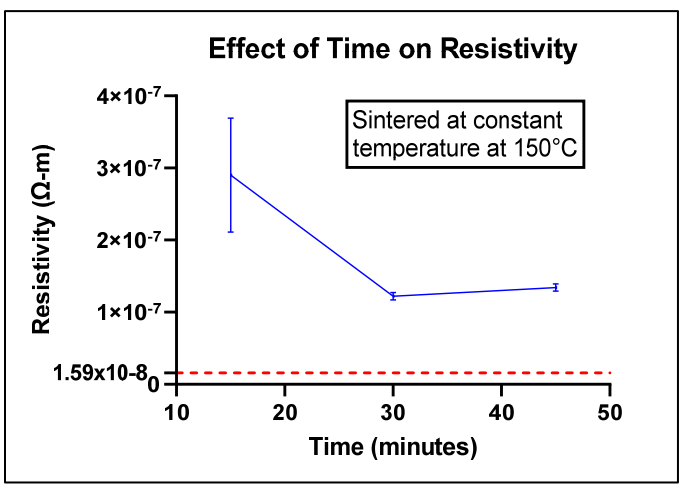


Fig. 16. Effect of Sintering Time on Electrical Resistance

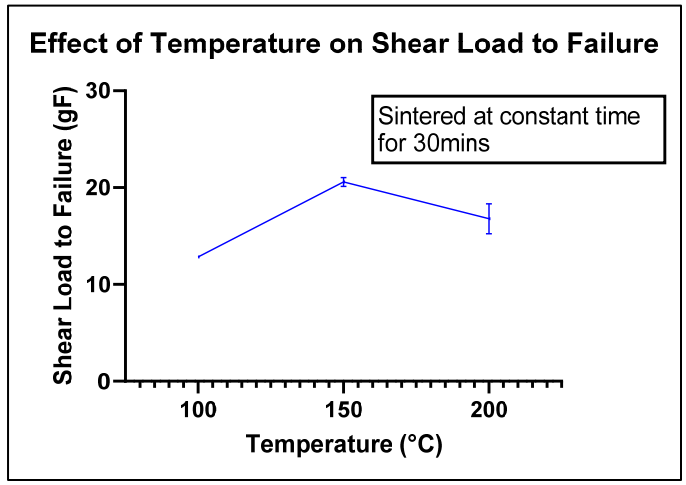


Fig. 17. Effect of Sintering Temperature on Shear Load

Effect of sintering post-print parameters on the mechanical property, shear load to failure, was also quantified. Looking back at the results shown in Fig. 15 and Fig. 16, sintering time was kept constant for 30 mins for the shear load to failure test. Multiple samples were tested for repeatability. Fig. 17 shows the graph of shear load to failure that was done on different samples sintered at 100°C, 150°C, and 200°C. Initial increase the shear load is to be expected because of the surfactant evaporation and increase in adhesion of silver Nanoparticles to the polyimide film. However, there is a decrease in shear load when the sample was sintered at 200°C. For this explanation, optical pictures were recorded for the samples. For the sample sintered at 200°C for 30 mins, it can be seen that the cracks start to develop that reduced the adhesion between the printed material and polyimide. Optical pictures at various temperatures along with zoomed-in view of the cracks are shown in Fig. 18, with a sheared line shown in Fig. 19. The shear test was done using Dage 2400PC Ball Shear Tester instrument. The shear tip was brought to the polyimide surface near the printed trace, and the shearing of line was done perpendicularly as shown in Fig. 19.

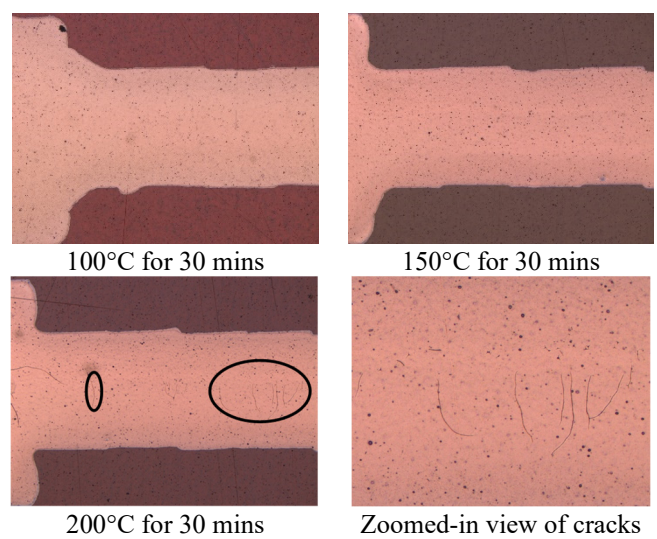


Fig. 18. Optical Pictures of Printed Track for Various Sintering Conditions

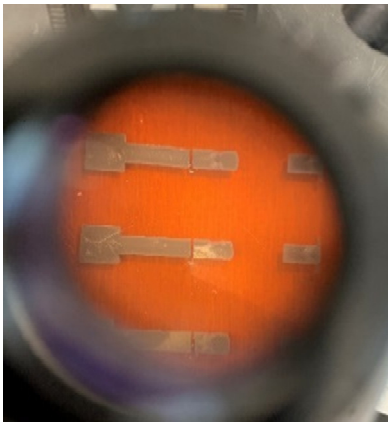
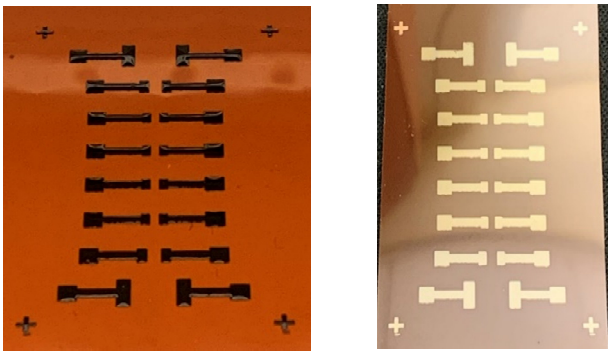


Fig. 19. A Close-up of Sheared Line

B. *SMD Discrete Attachment*

In this section final test design with SMD footprint was printed. The print before sintering is shown in Fig. 20 (left), and after sintering at 150°C for 30 mins is shown in Fig. 20 (right). Fig. 21 shows an optical picture of the defined CAD dimension against the actual measured. The 110 µm difference is to be expected because of the resolution of 42 pL print head.



Before sintering After sintering

Fig. 20. Before and After Sintering Pictures of Printed Tracks

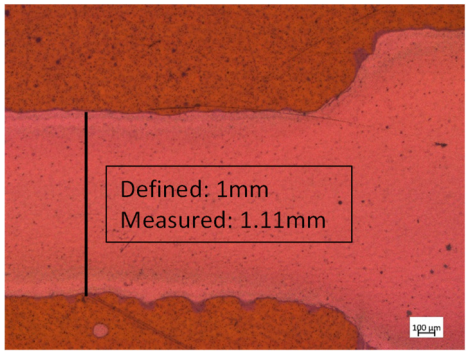


Fig. 21. Dimension of Printed Track

For the height of the printed design, a White Light interferometry scan was performed. It is an optical non-contact method that uses a beam-splitter to divide the light source into two beams: from the sample and from the known reference mirror. These two beams when mixed creates an interference pattern consisting of intensity that is related to the surface height. Fig. 22 shows the contour of IJ printed ink with a surface profile along a perpendicular line in y-direction for height measurement. The height of printed line comes to be about 1µm, as can be seen in Fig. 23.

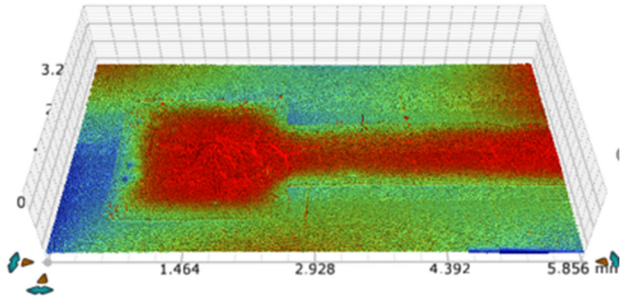


Fig. 22. 3D Surface Profile of the Printed Track

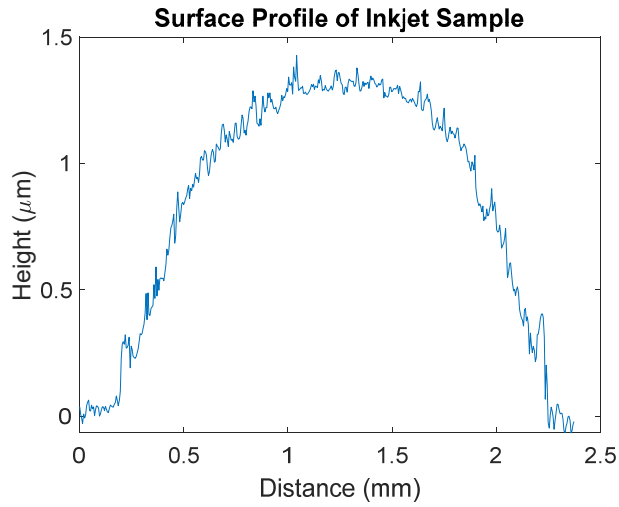


Fig. 23. Profile of Inkjet Printed Track

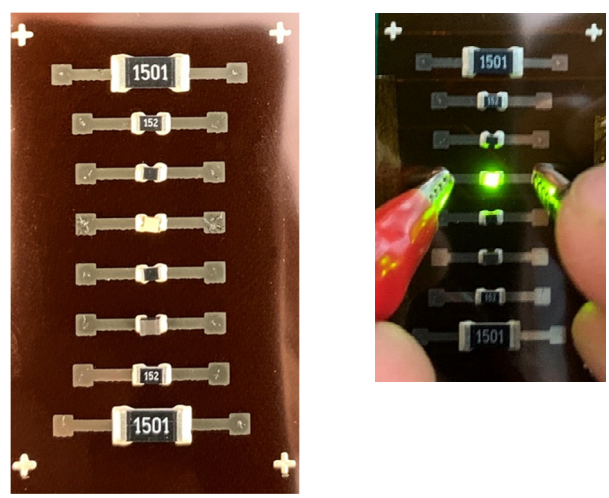


Fig. 24. SMD Attachment on Inkjet Printed Track, with lit LED (right)

Next, the ECA was dispensed on the silver printed SMD footprint using the DW printer. The process parameters used to dispense the ECA were optimized in another study. After the ECA dispensing, the SMD components were mounted using a semi-automatic pick-and-place instrument. A SMD LED was

also mounted to display the electrical connectivity. The whole assembly was cured in a thermal conventional oven using the manufacturer recommended condition of 150°C for 30 mins. Fig. 24 (left) shows the picture of mounted components and Fig. 24 (right) shows the lit LED when supplied with power

RLC measurements are carried out at various frequency levels for the attached discrete components. Fig. 25 shows the frequency variation of the SMD resistor, capacitor, and inductor from frequency of 10 kHz to 500 kHz. The instrument used is Agilent 4192A LF Impedance Analyzer. In all the three cases, there is negligible difference in the measured values as compared to the rated value of the component, however, in the case of resistor, there is decrease in the value at higher frequency, which is attributed to the parasitic capacitance that becomes dominant at 500 kHz.

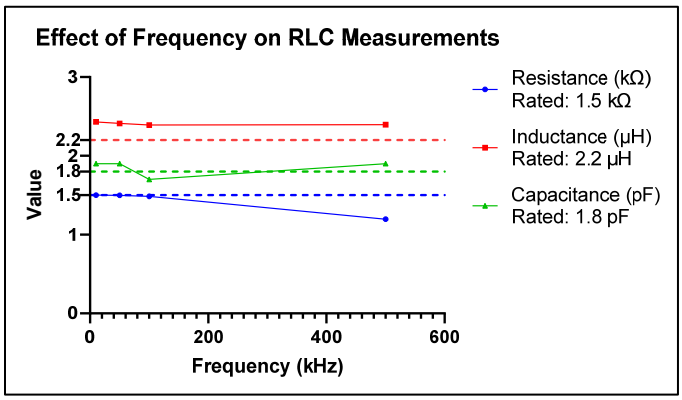


Fig. 25. Frequency Variation of Mounted SMDs with ECA and Inkjet Printed Pads

C. *Statistical Study*

A statistical study was performed to measure the variability in IJ print process. The technique used in this study is control chart. Control charting is a process that measures the variance in a process over time. It consists of user specification limits, control limits that are calculated based on means, standard deviation and number of sub-groups. In addition to the limits, it also consists of means of the sub-groups as well. Based on the number of sub-groups and the size, different types of control charts are available. The control limits are calculated using the following equations.

$$UCL = \left(\frac{\sum_{i=1}^{n} \bar{X}_i}{n} \right) + A \left(\frac{\sum_{i=1}^{n} \bar{s}_i}{n} \right) \quad (1)$$

$$LCL = \left(\frac{\sum_{i=1}^{n} \bar{X}_i}{n} \right) - A \left(\frac{\sum_{i=1}^{n} \bar{s}_i}{n} \right) \quad (2)$$

Where,

$$s = \sqrt{\frac{\left(X_i - \bar{X}\right)^2}{k-1}} \quad (3)$$

Here,

$\bar{X}_i$	Average for i[th] sub-group
s_i	Standard deviation for i[th] sub-group
n	Number of sub-groups
A	Control chart constant based on sub-group size
k	Sub-group size
UCL	Upper Control Limit
LCL	Lower Control limit

For the statistical variability, 1000 drops are printed at a specific diameter of 500 μm. This value was decided so to have a target value instead of printing a pixel drop. Fig. 26 shows the pictures of the printed drops. Using the equations mentioned above and assuming the USL and LSL of ±5% tolerance of defined diameter, mean variation in the diameter is plotted in Fig. 27. Over the range of 16 sub-groups of 1000 drops, it is clear that the mean variance (depicted by the black line) stays under both the limits, user specified limits and control limits.

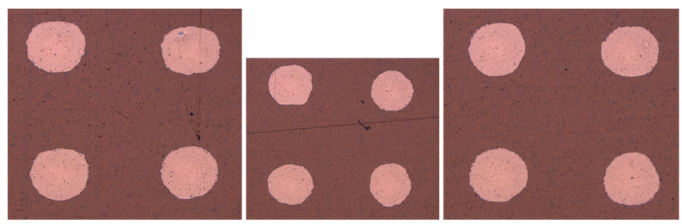


Fig. 26. Printed Inkjet Drops of 500 μm

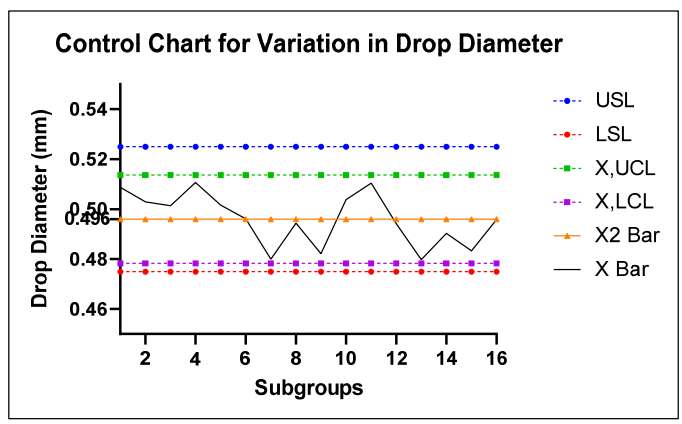


Fig. 27. Control Chart for Statistical Variance in Inkjet Process

Apart from control charting, statistical indices such as C_p – C_{pk} are also calculated, along with the distribution of the drop diameters. Fig. 28 shows the distribution with mean and standard deviation in the printed drops relative to the target value of the drop of 500 μm. USL and LSL are the user specification limits of ±5% of the target value. C_p – C_{pk} values are calculated using the following equations.

$$C_p = \frac{USL - LSL}{6\hat{\sigma}} \quad (4)$$

$$C_{pk} = \min\left(\frac{USL - \bar{x}}{3\hat{\sigma}}, \frac{\bar{x} - LSL}{3\hat{\sigma}}\right) \quad (5)$$

Here,

$\bar{x}$	Means
$\hat{\sigma}$	Standard deviation
USL	Upper Specification Limit (+5% of Target)
LSL	Lower Specification Limit (-5% of Target)

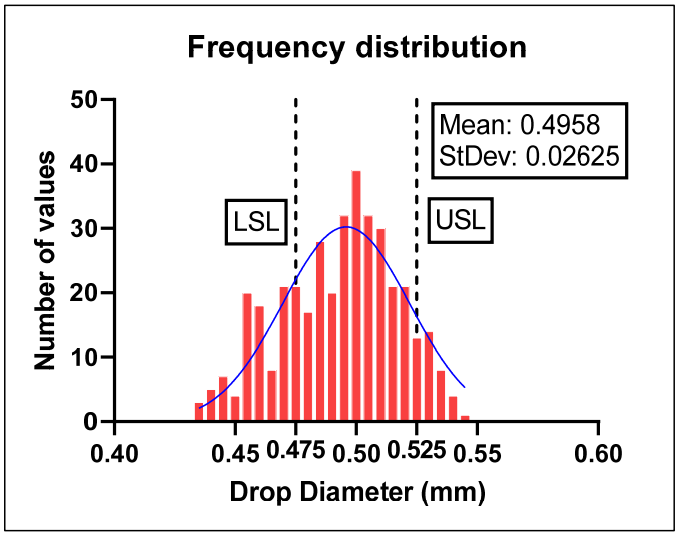


Fig. 28. Drop Distribution

IV. Summary and Conclusions

Direct write printing techniques such as IJ and micro-dispensing plays a prominent role in printed electronics for their high throughput allowing for rapid manufacturing of low-end electronic devices. However, both of the techniques requires a good understanding of the complete process for a fine print morphology. In the case of IJ, droplet formation and optimization of waveform for the print head used comes to be quite significant, while for DW micro-dispensing, timing of the material flow can affect the print greatly. In this paper, both of the techniques are utilized to achieve additive interconnections between a SMD and the substrate. First, IJ is used to print the trace and SMD footprint, followed by dispensing of conductive adhesive on just the footprint using DW micro-dispensing, and finally mounting of SMD components using semi-automatic pick-and-place machine. Substantial part of this paper is focused on IJ and process optimization for it. In IJ printing, waveform parameters tend to be more important since it affects the droplet speed, size, and shape. The first section of this paper discussed the effect of various waveform parameters on the droplet formation. Parameters taken into consideration are voltage, pulse width timing, phase width, and one reliability parameter, long term stability. After the finalization of waveform parameters, effect of post-print parameters such as sintering temperature and time is discussed on the electrical and mechanical properties of the printed feature. At lower temperatures and times, the printed feature was partially sintered that resulted in high resistance. There was negligible difference in the resistance when the temperature went above 150°C. Similar observation was made for sintering time. For constant temperature of 150°C, there was negligible difference between time of 30mins and 45mins. However, in the case of shear load to failure, sintering at temperatures above 150°C, keeping the time constant, cracks start to develop that degrades the print feature and its adhesion to the substrate, that is evident in Fig. 18. With the process parameters taken care of, SMD components are mounted next. For verification of electrical connectivity, a green LED was also mounted and shown. Process parameters for dispensing of ECA is not discussed in this paper. At the end, a statistical study is done. Control chart is plotted that shows the variation in IJ printing of around 500-600 drops over time. The variation stayed under the user specified limits that considers the ±4% variance from the defined drop diameter, and the control limits as well that are calculated based on the sub-group size, mean, and standard deviation. With this study, authors aim to quantify some of the process parameters and their effect on the droplet formation from IJ printing for fine print quality, and to achieve additive interconnections between the COTS SMD components and a flexible substrate.

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